

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)			higher temperature (1) higher rate is due to more successful collisions per second / greater frequency of successful collisions / more particles having required activation energy (1) at higher temperature particles have more (kinetic) energy / move faster so more of the collisions that occur are successful (1) larger surface area means that more particles are able to be involved in collisions (1)	4			4		4
	(b)			either of following • equal volume of gas produced in both experiments • graphs level off at the same volume in both experiments			1	1		1
				Question 5 total	4	0	1	5	0	5